SubSync: An Automated Web-Based Platform for Enhancing Community Engagement in Residential Subdivisions

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Chapter 1

INTRODUCTION

Background of the Study

The rapid advancement of information technology has significantly transformed how communities manage information, communication, and service delivery throughout generations. Digital platforms are increasingly used to improve transparency, efficiency, and collaboration within organizations and communities. In residential subdivisions, these technologies can help manage announcements, and community concerns more effectively. However, many subdivisions still rely on traditional methods such as paper-based records, bulletin board announcements, and manual communication. These methods often result in delays, miscommunication, and limited transparency for residents.

The transition towards technology-based community monitoring systems has been extremely effective in supporting local planning and decision-making processes. A good example is the Community-Based Monitoring System (CBMS), which highlights how local data can be useful in helping local governments identify specific needs in the communities they serve (Reyes & Mandap, 2022). Similarly, intelligent community management models emphasize the use of information technology and communication networks to improve governance and operational efficiency. These systems integrate communication, monitoring, and data management to support better community services (Tang, 2022). As communities grow, the use of digital systems becomes increasingly important for efficient administration.

Digital reporting platforms also play an important role in encouraging community participation. These systems allow residents to submit complaints, concerns, and suggestions directly to administrators. Such platforms enable faster monitoring and response to issues within the community (Anggara et al., 2023). In addition, interactive mapping technologies improve usability by visually presenting geographic information. This allows users to easily locate facilities and identify reported issues within a subdivision (Bartling et al., 2021).

Web-based information systems have also been widely used to manage resident data and improve service delivery. These systems simplify the storage, access, and management of community information compared to traditional methods. They also enhance reliability, usability, and efficiency in handling administrative tasks (Melendres & Aranda, 2022). Despite these technological developments, many residential subdivisions still lack integrated digital management platforms. As a result, residents may experience limited access to information and reduced engagement with community management.

To address these challenges, this study proposes the development of SubSync: An Automated Web-Based Platform for Enhancing Community Engagement in Residential Subdivisions. The system aims to improve transparency, communication, and participation within residential communities. It integrates a communication tools, issue reporting, recommendations, and interactive mapping features. These features are designed to provide a centralized platform for residents and administrators. Through this system, the study seeks to enhance community engagement and improve subdivision management.

Objectives of the Study

The study aims to design and develop a web-based community management platform for residential subdivisions that enhances resident engagement through, communication, issue reporting, recommendations, and interactive mapping

Specifically, the study aims to:

1. Design and develop a community management system featuring:
    1.1. An integrated communication module that enables direct messaging between resident and administrator. It allows administrators to post announcements, updates, and subdivision events.
    1.2. An issue reporting system that allows residents to submit complaints and concerns while enabling administrators to efficiently monitor and respond to subdivision-related issues.
    1.3. A recommendation feature that allows residents to suggest improvements and initiatives for the overall development of the subdivision.
    1.4. A dynamic mapping feature that provides a visual representation of the subdivision layout, enabling residents and administrators to locate facilities, emergency tracking, monitor reported issues geographically, and support planning and decision-making through interactive map-based data visualization.

2. Test the functionality of the features mentioned above
3. Evaluate the system using ISO 25010 standardize evaluation tool in terms of

3.1. Functional Suitability

3.2. Performance Efficiency

3.3. Usability

3.4. Compatibility

3.5. Reliability

3.6. Security

3.7. Maintainability

3.8. Portability

Significance of the Study

The study SubSync: An Automated Web-Based Platform for Enhancing Community Engagement in Residential Subdivisions seeks to improve the management of subdivision operations, communication, and community participation through a centralized digital platform. The system is designed to streamline processes such as issue reporting, announcements, and community recommendations while providing an interactive mapping feature for better visualization of subdivision facilities and concerns. By integrating these functions into a single web-based platform, the system aims to enhance transparency, efficiency, and accessibility for both residents and administrators. Furthermore, it supports improved decision-making and community collaboration within residential subdivisions. This system is intended to deliver significant benefits to the following stakeholders.

Subdivision Residents. Residents will benefit from improved access to announcements, and community updates. The platform will also allow them to actively participate in community improvement through issue reporting and recommendations.

Subdivision Administrators. Administrators will gain a centralized system that simplifies communication, and issue management, enabling them to respond more efficiently to residents' concerns.

Homeowners Associations. The system will improve transparency and strengthen relationships with residents by providing clear and accessible administrative information, announcements, community concerns, and recommendations.

Researchers. The researchers will enhance their skills in developing web-based systems, including database management, system integration, and user interface design. Through the development of SubSync, they will gain hands-on experience in implementing features such as, communication modules, issue reporting, and interactive mapping. This study also allows them to apply their academic knowledge to a real-world community management system.

Future Researchers. This study may serve as a reference for future research related to community management systems, digital governance platforms, and smart community technologies.

Scope and Limitation of the Study

This study focuses on the design, development, and evaluation of SubSync: An Automated Web-Based Platform for Enhancing Community Engagement in Residential

Subdivisions. The system is intended to be implemented within selected residential subdivisions to improve community management, communication, and participation. It includes key features such integrated communication tool, issue reporting, recommendation submission, and dynamic mapping. The platform is web-based and accessible through desktop and mobile browsers, allowing both residents and administrators to interact efficiently. The study also involves testing and evaluating the system using the ISO 25010 software quality standards, including functionality suitability, performance efficiency, usability, compatibility, reliability, security, maintainability, and portability.

Despite its contributions, the study has several limitations that must be considered. The system will be tested only within a selected subdivision or group of users, which may limit its applicability to other communities. The platform relies on stable internet connectivity, making it less effective in areas with poor or unstable network access. The mapping feature only supports basic visualization and does not include advanced geospatial analysis, while other services such as security monitoring and emergency response systems are covered.

Definition of Terms

For better comprehension of the study, the following terms are hereby operationally defined in both conceptual and operational contexts.

Administrator. Conceptually, an administrator is a user responsible for managing system operations and overseeing activities (Melendres & Aranda, 2022).

As applied in this study, administrators respond to issues, post announcements, and monitor overall system usage within the SubSync platform.

Centralized System. Conceptually, a centralized system is a structure where all data and processes are managed in a single location or platform (Tang, 2022).

As applied in this study, SubSync serves as a unified system where all subdivision data, communication, and transactions are stored and managed.

Communication Module. Conceptually, a communication module is a system component that facilitates the exchange of information between users (Tang, 2022).

As applied in this study, it enables direct messaging between residents and administrators and allows posting of announcements and updates.

Community Engagement. Conceptually, community engagement refers to the active participation and involvement of individuals in activities that promote collaboration and shared decision-making within a community (Anggara et al., 2023).

As applied in this study, it refers to the participation of subdivision residents in submitting concerns, recommendations, and interacting with administrators through the SubSync platform.

Dynamic Mapping. Conceptually, dynamic mapping refers to interactive digital maps that display spatial information and update based on user interaction or data changes (Bartling et al., 2021).

As applied in this study, it visually represents subdivision layouts, facilities, and reported issues to support navigation and decision-making.

Issue Reporting System. Conceptually, an issue reporting system allows users to submit and track complaints or concerns within an organization or community (Melendres & Aranda, 2022).

As applied in this study, residents can report subdivision-related problems, and administrators can monitor and respond through SubSync.

Recommendation System. Conceptually, a recommendation system allows users to provide suggestions for improvements or actions within a system or organization (Anggara et al., 2023).

As applied in this study, residents can submit ideas and proposals to improve subdivision services and facilities.

Resident. Conceptually, a resident is an individual living within a specific community or area (Reyes & Mandap, 2022).

As applied in this study, residents are users who access SubSync to receive updates, and participate in community activities.

Subdivision Management System. Conceptually, a subdivision management system is a digital solution designed to manage operations, communication, and services within a residential community (Tang, 2022).

As applied in this study, SubSync functions as a subdivision management system communication, issue reporting, and mapping features.

Web-Based Platform. Conceptually, a web-based platform is a system that operates through a web browser, allowing users to access services over the internet without installing software (Melendres & Aranda, 2022).

As applied in this study, it refers to SubSync, which enables residents and administrators to access features such as communication, and issue reporting via browsers.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents related literature and studies from foreign and local sources that are relevant to the present study. It covers existing systems, technologies, and research related to community management systems, issue reporting systems, communication platforms, geospatial mapping, and web-based dashboard systems. These studies provide a basis for understanding current technologies and identifying gaps and limitations that support the development of the proposed system.

Community Management Systems for Residential Communities

Web-based community management systems provide centralized platforms for storing data, facilitating communication, and coordinating services within defined groups. Research on digital platforms that support community assessment and governance emphasizes that organizational factors and system-wide design standards significantly influence the effectiveness of implementation and long-term community health (Luo & Chen, 2024). These systems help users interact, share information, and collaborate within a digital environment, acting as a backbone for community operations. Features such as user profiles, member directories, and structured service modules make digital community platforms instrumental for building transparency and cohesion among residents.

In the Philippine setting, community platforms like Webyu and barangay management information systems illustrate how localized solutions improve grassroots

governance. Studies have shown that web-based barangay systems enhance administrative efficiency, reduce errors, and enable residents to access services without relying on manual procedures (Gasmido et al., 2025). Similarly, community e-services platforms integrate multiple functions — including resident data, notifications, and service requests — demonstrating that comprehensive community management systems can strengthen transparency and responsiveness in local residential settings (Ching & Orozco, 2025). These examples affirm the relevance of centralized community management systems in enhancing operational efficiency and resident participation.

Issue Reporting and Complaint Management Systems

Issue reporting and complaint management systems streamline how communities document concerns and enable administrators to respond efficiently. Foreign studies on web-based public complaint platforms illustrate that digital reporting mechanisms enhance transparency and responsiveness by offering structured channels for issue submission, categorization, and monitoring (Shofiana et al., 2025). These platforms replace manual or paper-based systems, helping officials manage records more effectively and analyze trends in public feedback to improve services.

In the Philippines, digital reporting tools are increasingly adopted for disaster response and barangay governance. For instance, barangay disaster relief systems include modules that capture resident reports and feedback, accelerating emergency coordination and follow-up actions (Baylon et al., 2026). Similarly, localized online reporting tools such

as community complaint mechanisms have been shown to increase accessibility for residents and reduce response times for officials tasked with addressing public concerns (Ningrum et al., 2025). These studies support the idea that web-based issue reporting systems are essential components of modern community platforms, enhancing accountability and facilitating more responsive governance.

Communication and Announcement Systems

Communication and announcement systems facilitate real-time information exchange between administrators and residents. Research indicates that digital channels such as automated alerts, messaging modules, and bulletin boards increase participation, reduce misinformation, and help communities stay informed about relevant issues (Systematic analysis of digital tools for citizen participation, 2024). These systems enable administrators to broadcast updates to many users simultaneously, improving access to local news, meeting notices, and service changes.

Philippine studies also highlight the value of integrated communication features in local governance systems. Platforms like Webyu and LOKAL Pilipinas include announcement and alert modules that streamline how updates are disseminated to residents, fostering improved connectivity and engagement (Gasmido et al., 2025; Ching & Orozco, 2025). Research on communication skill enhancement among barangay officials further shows that effective use of digital communication tools enhances information flow and supports participatory governance (Ferranco et al., 2025). This body of research

underscores the importance of adaptable communication systems in web-based platforms to keep communities informed and engaged.

Web-Based Administrative Dashboard Systems

Web-based administrative dashboards unify disparate community data into a cohesive visual interface that supports monitoring, evaluation, and decision-making. Research on digital civic dashboards underlines their role in presenting complex datasets — such as service metrics, finance indicators, and resident feedback — through intuitive graphs, charts, and summaries (Pluto-Kossakowska et al., 2025). Clear visualizations reduce information overload and improve administrators' ability to access relevant insights quickly, fostering evidence-based governance.

Locally, digital dashboard solutions have been developed to improve transparency, data access, and service quality in various governance contexts. For example, dynamic web-based information systems developed in local government units provide consolidated dashboards for financial records, service requests, and community profiles, enabling officials to monitor performance and manage resources more efficiently (Albano, 2024). Recent evaluations of specialized local management systems, such as the Barangay Blotter and Clearance System in Cagayan, demonstrate that automating day-to-day transactions through computerized modules significantly enhances the efficiency of administrative operations and the overall experience for residents (Alvarez et al., 2023). These local

examples affirm the value of administrative digitalization in fostering more responsive and organized community management.

Table 1

Matrix of Related Literatures and Studies

RRL	**Features**					
System / RRL	Community Mgmt	Community Mgmt	Issue Reporting	Communication	Geospatial Mapping	Web-Based Dashboard
Community Management Information System	✓	✓	✓	✓	×	✓
Complaint Management System	×	×	✓	✓	×	✓
GIS-Based Community Mapping System	✓	×	×	×	✓	✓
Web-Based Cooperative Management System	✓	✓	×	✓	×	✓
Smart Barangay Management System	✓	✓	✓	✓	✓	✓
Online Payment and Billing System	×	✓	×	✓	×	✓
Maintenance Request Management System	×	×	✓	✓	×	✓
Community Communication Portal	✓	×	×	✓	×	✓
Integrated Community Service System	✓	✓	✓	✓	×	✓
Digital Resident Information System	✓	×	×	✓	×	✓
Issue Tracking and Resolution System	×	×	×	✓	×	✓
Barangay Information Management System	✓	✓	✓	✓	×	✓
GIS-Based Disaster Risk Mapping System	×	×	×	×	✓	✓
Online Announcement and Notification System	×	×	×	✓	×	✓

Web-Based Administrative Dashboard System	✓	✓	×	×	×	✓

Synthesis

The reviewed related literature and studies show that community management systems have significantly improved the way residential communities, organizations, and cooperatives manage their operations through digital platforms. These systems commonly include community or member management, issue reporting, communication and announcement systems, and web-based dashboards. The literature emphasizes that digital systems improve operational efficiency, transparency, data accuracy, and communication between administrators and residents.

Communication platforms also play an important role in disseminating announcements, reminders, and notifications, which helps improve community participation and coordination. Furthermore, several studies highlight the importance of web-based dashboards in supporting planning, decision-making, and data visualization, helping administrators monitor transactions, reports, and system activities through graphical presentations.

However, the literature also reveals that most existing systems focus only on specific features and do not integrate all functionalities into a single platform. This indicates a gap in existing systems, particularly in integrating community management, ,issue reporting, communication systems, and dashboard monitoring into one unified

system. Therefore, the development of the proposed system aims to address this gap by providing an integrated community management platform that streamlines all subdivision operations into one accessible interface.

Chapter 3

DESIGN AND METHODOLOGY

Research Design

The study employs a Descriptive-Developmental Research Design to guide the design, development, and evaluation of SubSync: An Automated Web-Based Platform for Enhancing Community Engagement in Residential Subdivisions. This research design is appropriate for the study because it involves both an examination of the subdivision's existing conditions and the development of a system that addresses identified operational problems. This ensures that the system developed is aligned with the community's needs and administrative operations.

The descriptive component of the research design is used to examine and describe the current manual processes within the residential subdivision. Descriptive research is defined as a method that systematically describes a population, situation, or phenomenon without manipulating variables, focusing on existing conditions (Osio et al., 2026). In this study, it includes the management of resident records, dissemination of announcements, and the handling of community issue reports. This phase focuses on identifying and understanding operational issues such as delays in communication, data redundancy in paper-based ledgers, and the lack of a centralized data management system (Pinero, 2024). The results of this phase serve as the basis for determining the functional requirements of the proposed platform.

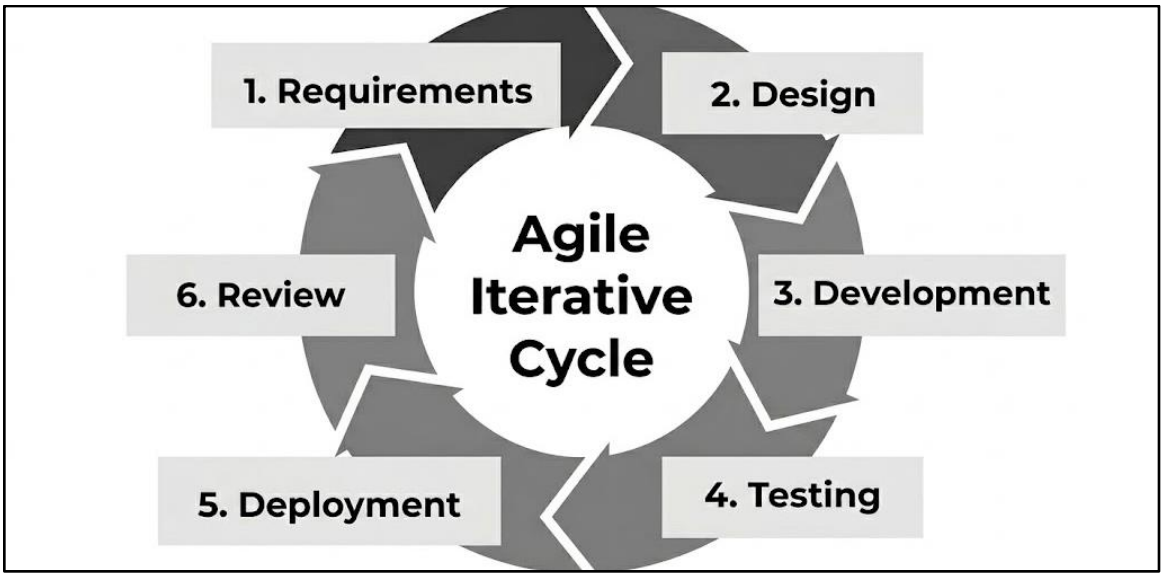
The developmental component of the research design is applied in the design and development of SubSync, which involves the creation of a web-based platform for administrators and residents. Development in system and software research refers to the process of designing, building, and iteratively improving software systems to solve real-world problems and meet user requirements (Perera, 2025). The system integrates key modules such as resident management, an automated announcement system, and an issue reporting module. The development process follows an Agile Methodology, an iterative approach involving continuous cycles of planning, design, development, and testing to ensure that the system effectively addresses identified problems and meets the operational requirements of the subdivision (Ilustrisimo, 2026).

Through this research design, SubSync is developed based on the actual needs and operational challenges of the subdivision stakeholders, ensuring the system is grounded in real-world conditions and accurately reflects the existing workflow. By combining both descriptive analysis and developmental system design, the study bridges the gap between manual processes and digital solutions. This approach results in an integrated and centralized platform that improves data accuracy, enhances operational efficiency, and streamlines the overall management of residential activities, which is then evaluated using the software quality standards of ISO/IEC 25010:2011 (Dandan, 2026).

System Development Life Cycle

Figure 1

Agile Development Model



Agile development is an iterative and incremental software development approach that focuses on continuous improvement and flexibility through repeated cycles (Žužek et al., 2021). It is cyclical in nature, meaning that each iteration includes planning, design, development, and testing activities to ensure that each functional increment meets quality standards before integration (Vreuls et al., 2023). For SubSync: An Automated Web-Based Platform for Enhancing Community Engagement in Residential Subdivisions, the development process follows this agile system development life cycle (SDLC) framework. This framework allows for iterative progress through planning, design, development, testing, deployment, and evaluation to address the need for a centralized, transparent, and efficient community management tool (Dingle, 2026). This methodology enables

researchers to refine key features—such as automated announcements—based on continuous feedback and incremental reviews of the system (Bastos, 2025).

The key phases include:

A. Requirements Analysis

In this phase, the researchers identify the functional requirements for a centralized system, such as replacing paper-based records with a digital platform for managing resident information, announcements, community concerns, and recommendations. This stage focuses on understanding the specific communication and management gaps within residential subdivisions to ensure that the platform facilitates effective service delivery.

B. Sprint Planning

The development process is divided into iterative cycles, prioritizing core modules like the integrated communication tool for resident-administrator messaging and the dynamic mapping feature for facility and emergency tracking. By breaking down the project into sprints, the researchers can focus on high-priority features first, ensuring a steady progression of functional system increments.

C. Design and Implementation

During each sprint, the technical architecture is developed using a web-based tech stack, focusing on creating intuitive dashboards that simplify resident information, community concerns, announcements, and recommendations for non-technical users. This involves translating the subdivision layout into a dynamic mapping system that supports better community navigation and decision-making through interactive data visualization.

D. Testing and Review

Each completed module, such as the issue reporting system or the recommendation feature, undergoes rigorous functionality testing to identify and fix bugs early in the process. This stage ensures that automated notifications and geospatial visualizations respond accurately to user inputs, maintaining the reliability and responsiveness of the platform.

E. Deployment and Evaluation

Following the integration of all modules including the announcement systems the platform is finalized for a group of selected users. The system is then formally evaluated using the ISO 25010 quality standards, specifically assessing its functional suitability, performance efficiency, and security to ensure it meets the needs of subdivision stakeholders and IT experts.

Research Respondents

The respondents of this study are the main users and evaluators of the SubSync platform. They were chosen because they are directly involved in subdivision activities and can provide reliable feedback about how the system works in terms of functionality, usability, and effectiveness. A total of 25 respondents participated in the study, consisting of the subdivision president/administrator, homeowners' association officers, residents, and IT experts or IT professionals.

The first group is the Subdivision Administrator and HOA President, who oversee the daily operations and governance of the residential community. This group was selected because they manage the overarching records and financial summaries of the subdivision. Their feedback focuses on how the web-based system helps improve administrative oversight, especially in terms of data organization and ease of management.

The second group is the Homeowners' Association (HOA) Officers, who assist in the implementation of community rules and the dissemination of information. They were selected because they are the ones handling specific administrative tasks and coordinating community announcements. Their feedback focuses on the speed and reliability of the platform's communication modules compared to traditional methods.

The third group is the Subdivision Residents, who are the registered homeowners and tenants within the community. They are the primary users of the system for services like viewing personal payment histories and submitting community reports. Their feedback

is important in knowing if the system is easy to use, helpful, and suitable for their daily needs as members of the community.

The fourth group is the IT Experts or IT Professionals, who have experience in system development and software evaluation. They help check the technical side of the system, including its performance, security, and overall quality. Their input ensures that the system is not only easy to use but also technically strong and reliable based on industry standards.

Overall, these four groups provide balanced feedback for the SubSync platform. Their responses help determine if the system is effective in terms of both user experience and technical performance. Table 2 shows the number of respondents and their distribution in terms of number and percentage.

Table 2

Research Respondents

Respondent Type	**No. of Respondents**	**Percentage**
Subdivision Administrator/President	1	4%
HOA Officers	10	28%
Subdivision Residents	10	28%
Respondent Type	**No. of Respondents**	**Percentage**
IT Experts/IT Professionals	5	40%
Total	25	100%

Sampling Technique

The researchers will employ a Purposive Sampling Technique, a prominent non-probability sampling method where participants are selected based on the researcher's judgment regarding which individuals will be most informative for the study's specific objectives. Unlike probability-based methods that rely on random selection, purposive sampling focuses on criterion-based selection, ensuring that each respondent possesses unique insights or experiences essential to the phenomenon under study. This approach is particularly effective in research where the validity of the results depends on the depth of the participants' first-hand knowledge rather than general representativeness.

For the evaluation of SubSync, this technique is appropriate as it allows the researchers to target individuals with direct, practical experience in subdivision living or administrative management. Selecting respondents such as homeowners and subdivision officers ensures that the feedback regarding the platform's functional suitability and usability is grounded in real-world administrative challenges. Furthermore, for the technical evaluation phase, the researchers specifically target IT Professionals and Software Experts who meet predefined inclusion criteria, such as prior experience in system analysis or software quality assessment. These experts possess the specialized technical competency required to audit a web-based platform's architecture and security protocols. By deliberately selecting both end-users and technical specialists, the study achieves a balanced and rigorous analysis of the system's performance based on established global standards like ISO/IEC 25010.

Data Gathering Instrument

The primary research instrument used in this study is a combination of qualitative and quantitative tools designed to capture both the functional requirements and the eventual quality of the system. For the initial phase, the researchers will utilize a Semi-Structured Interview Guide to gather in-depth insights from subdivision administrators and homeowners' association officers. This guide consists of open-ended questions focused on current manual workflows, common bottlenecks in payment tracking, and existing gaps in community communication. The flexibility of a semi-structured interview allows the researchers to ask follow-up questions to clarify complex administrative tasks that the "SubSync" platform must automate.

Following the development of the platform, a Software Quality Evaluation Questionnaire adapted from the ISO/IEC 25010 standard will be employed. This instrument is designed to quantitatively measure the system's performance across eight key characteristics: Functional Suitability, Performance Efficiency, Usability, Compatibility, Reliability, Security, Maintainability, and Portability. The questionnaire utilizes a 5-point Likert Scale, ranging from "Strongly Disagree" (1) to "Strongly Agree" (5), providing a standardized metric for both IT professionals and end-users to assess the platform’s effectiveness in a subdivision environment.

Data Gathering Procedure

The data gathering procedure for this study begins with formal coordination with the selected residential subdivision and a request for permission from the homeowners'

association or subdivision management to conduct the research. Once approval is granted, we will conduct semi-structured interviews with the subdivision administrators and staff to gather detailed information regarding their current manual management and communication processes. These interviews will be conducted through face-to-face sessions and online communication channels to ensure a comprehensive understanding of operational bottlenecks, such as manual payment tracking and bulletin-based announcements.

The responses from these interviews will be documented, organized, and analyzed to identify specific system requirements and guide the iterative development of the "SubSync" platform. The qualitative data gathered through these sessions will serve as the foundational basis for designing a system that accurately meets the operational needs of both residents and administrators. Furthermore, these collected insights will support the final system evaluation phase, where an ISO/IEC 25010-based instrument will be utilized to assess the platform's quality, performance, and overall effectiveness in enhancing community engagement.

Validity of Research Instrument

A semi-structured interview guide was developed to obtain detailed information regarding the current administrative and communication processes of the selected residential subdivision in relation to the development of SubSync: An Automated Web-Based Platform for Enhancing Community Engagement in Residential Subdivisions. The guide covers essential operational domains such as resident record management, post-

manual payment tracking, announcement dissemination, issue reporting, and facility mapping. The development of the interview questions was guided by the specific objectives of the study to ensure that the gathered data accurately reflects actual operational requirements and community needs. The instrument was reviewed by the researchers and refined under the guidance of the academic adviser to ensure clarity, relevance, and alignment with the study framework. This ensures that the interview guide is appropriate for eliciting comprehensive responses from the subdivision administrators and homeowners' association officers regarding existing workflows and community engagement challenges.

Reliability of the Research Instrument

To ensure consistency in data collection was ensured through the use of a standardized semi-structured interview guide administered to the secretary, encoder, and manager. All participants were provided with the same set of structured questions to maintain uniformity and allow systematic comparison of responses. Follow-up questions were utilized only when necessary to clarify responses and ensure completeness of the information gathered. The interviews were conducted following a consistent procedure, and all responses were properly documented through recording and note-taking to preserve accuracy. This process ensures that the data collected across all respondents remains consistent, dependable, and suitable for analysis.

Analysis / Treatment of Data

The data collected from the interviews are analyzed using thematic analysis to systematically interpret the responses of the subdivision administrators and officers. All responses are transcribed, examined, and organized to identify recurring patterns, significant ideas, and operational bottlenecks within the residential community. These patterns are categorized to represent core processes, such as the manual recording of dues and the traditional methods of posting announcements. The analysis focuses on revealing issues such as communication delays, limited transparency in financial tracking, and inefficiencies in responding to resident reports. The findings serve as the basis for defining the system requirements of SubSync, while the ISO/IEC 25010 standard will be utilized during the system evaluation phase to assess software quality attributes through weighted mean and descriptive statistics.

Ethical Consideration

Ethical standards are strictly observed throughout the conduct of the study to ensure the protection of all participants involved. Informed consent is obtained from the subdivision administrators, officers, and residents prior to the data gathering process, and they are fully informed of the study's purpose, scope, and the functionality of the SubSync platform. Participation is entirely voluntary, and respondents are allowed to withdraw at any stage without any form of penalty or consequence. Confidentiality is maintained in accordance with the Data Privacy Act, ensuring that no personally identifiable information

are disclosed in the study. All collected data are used solely for academic purposes and are securely stored to protect participant privacy and subdivision data integrity.

Operational Framework

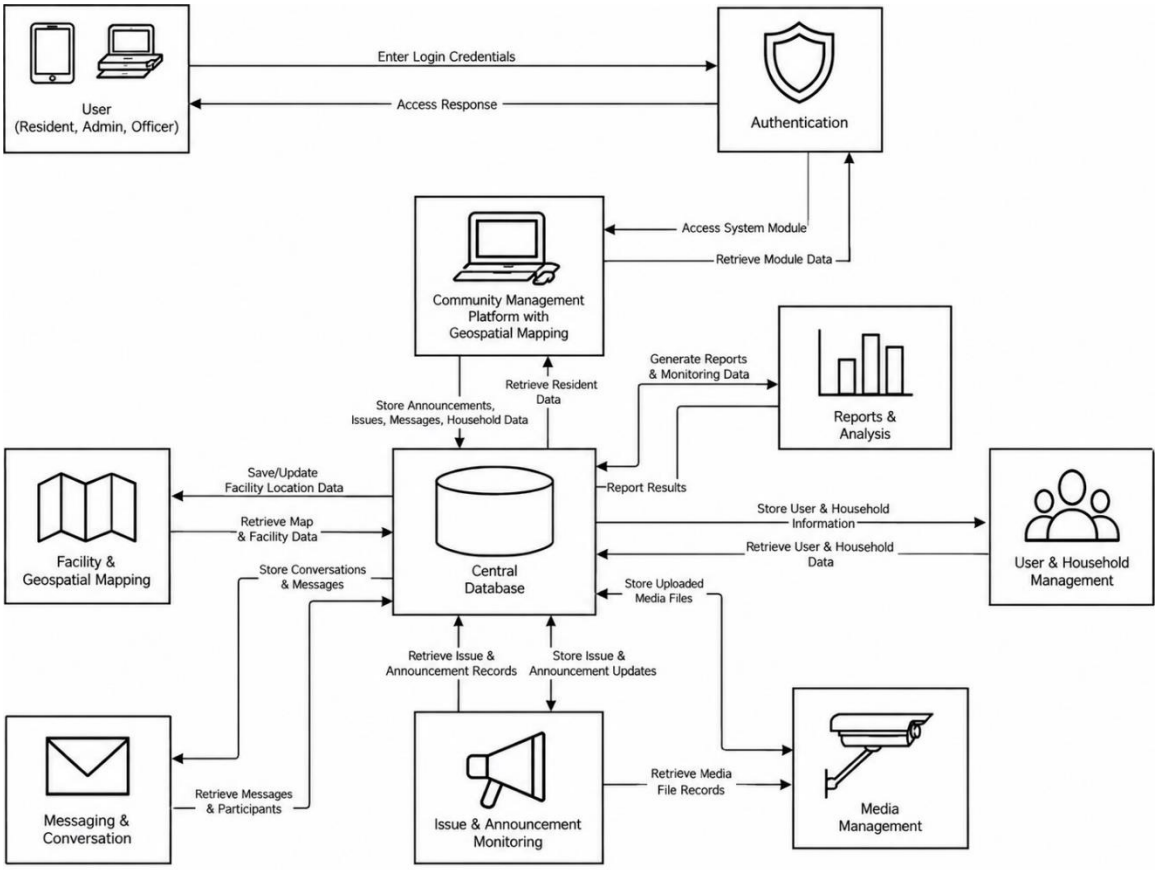
An operational framework refers to a structured representation that illustrates how different components of a system interact and work together to achieve its intended functions and objectives. In information systems development, frameworks are used to organize system components, define relationships among processes, and guide the flow of data within a system to ensure proper system functionality and alignment with organizational goals (Meneses & Varajão, 2022).

Figure 2 presents the Operational Framework of the SubSync system, which illustrates the interaction among the users, system modules, central database, and administrative processes. The framework shows that the User or Resident, Admin, and Officer first access the system by entering login credentials through the Authentication module to ensure secure and authorized entry. Once an access response is received, the users can interact with the SubSync: An Automated Web-Based Platform for Enhancing Community Engagement in Residential Subdivisions, which serves as the core interface for accessing system modules and managing essential information such as resident profiles, announcements, issues, recommendations, messages, and mapping data.

The system processes and manages community data by communicating with the Central Database for the systematic storage and retrieval of information across various specialized modules. Specifically, the User and Household Management component manages resident and household information, while the Facility & Geospatial Mapping module manages the location data of subdivision facilities and other relevant areas. Furthermore, the Issue & Announcement Monitoring and Messaging & Conversation components support community engagement by managing reported concerns, announcements, updates, and communication between users. The Media Management component handles uploaded media files, while the Reports and Analysis component generates organized reports and monitoring data based on the collected information. These components work together to support effective decision-making, communication, issue management, and overall administration of the residential subdivision.

Figure 2

OperationalFramework



Context Diagram

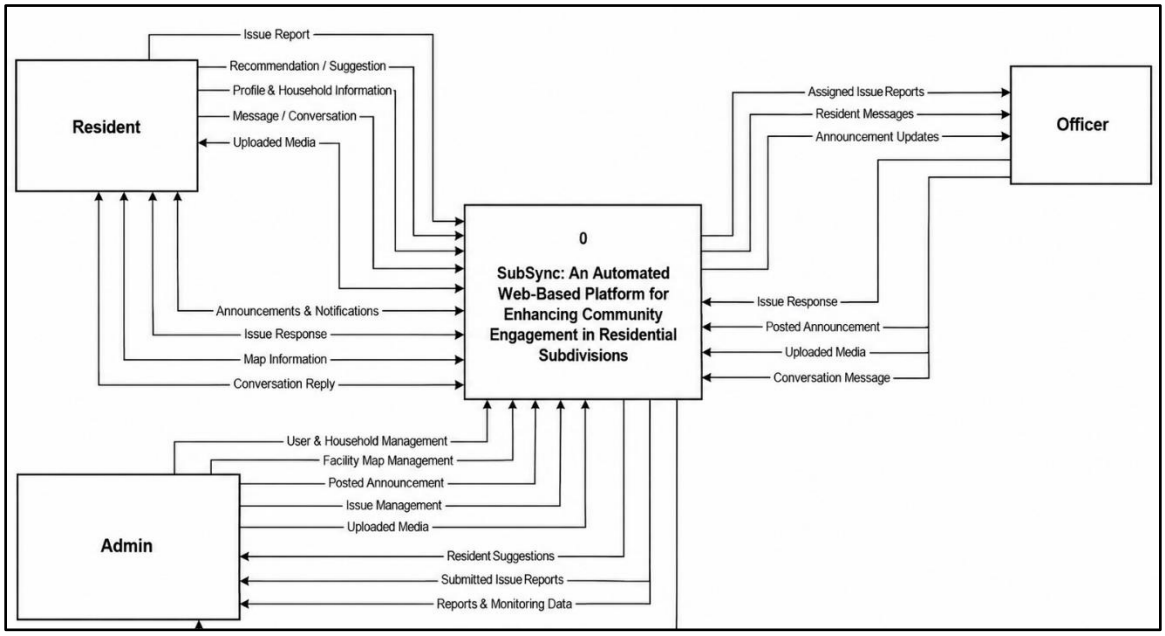
A context diagram, also known as a Level 0 Data Flow Diagram (DFD), represents the entire system as a single process to demonstrate the scope of the system and its interactions with external entities. This diagram illustrates the boundaries of the system and how information is exchanged between the platform and its primary users to ensure

streamlined subdivision operations. As illustrated in the SubSync system, identified as the SubSync: An Automated Web-Based Platform for Enhancing Community Engagement in Residential Subdivisions, acts as the central hub for data exchange among three primary entities: the Resident, Admin, and Officer.

The Resident entity provides inputs such as issue reports, recommendations, profile and household information, uploaded media, and conversation replies to the system. In return, the system provides the resident with announcements, notifications, issue response updates, facility map information, and conversation messages. The Admin serves as the primary controller of the system, providing essential data such as user and household management details, facility map management, posted announcements, uploaded media, and issue management. The system provides the admin with resident suggestions, submitted issue reports, reports and monitoring data, and system alerts and notifications. Finally, the Officer interacts with the system to manage community concerns and communication. The system sends assigned issue reports, resident messages, and announcement updates to the officer, who in turn provides issue responses, posted announcements, uploaded media, and conversation messages back to the platform. By defining these high-level data flows, the context diagram ensures that all stakeholders have a clear understanding of the system's functional boundaries and the specific information required for effective community management.

Figure 3

Context Diagram



Data Flow Diagram

The Data Flow Diagram (DFD) provides a detailed visualization of the internal processes, data stores, and information pathways within the SubSync platform. As shown in the system, it is broken down into six core functional processes that manage the flow of information between the Resident, Admin, and Officer entities. The User & Household Management (1.0) process handles account creation, profile updates, and household information, storing user data in the D1 User Accounts data store and issuing system alerts or confirmations back to the users. The Announcement Management (2.0) process enables admins and officers to post or manage announcements, which are stored in D2

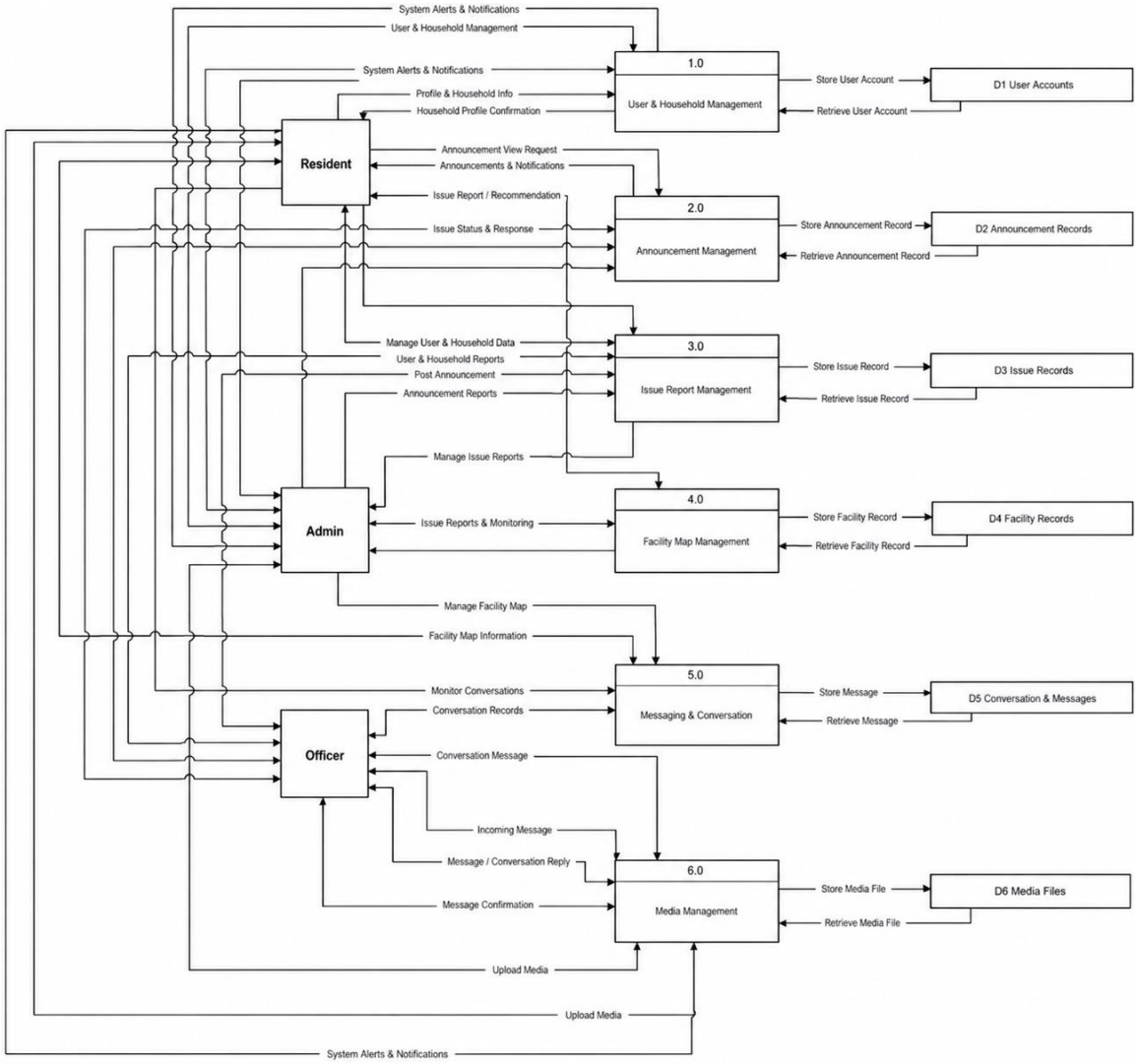
Announcement Records and retrieved when residents request to view current community updates.

Community concerns are handled through the Issue Report Management (3.0) process, where resident reports and recommendations are stored in D3 Issue Records. These reports can be managed and monitored by administrators and officers, who provide responses and status updates to residents. For physical community assets, the Facility Map Management (4.0) process stores location data in D4 Facility Records, allowing users to request and receive facility and map information.

Engagement and documentation are supported by the final two modules. The Messaging & Conversation (5.0) process manages direct communication by storing interaction data in D5 Conversation & Messages, facilitating the exchange of incoming and outgoing messages among residents, officers, and administrators. Finally, the Media Management (6.0) process serves as the repository for visual documentation, storing uploaded files in D6 Media Files and allowing users to upload and retrieve media files. Collectively, these processes provide a streamlined, data-driven approach to subdivision management by improving communication, issue reporting, information management, and community engagement through a centralized digital platform.

Figure 4

Data Flow Diagram



Entity Relationship Diagram

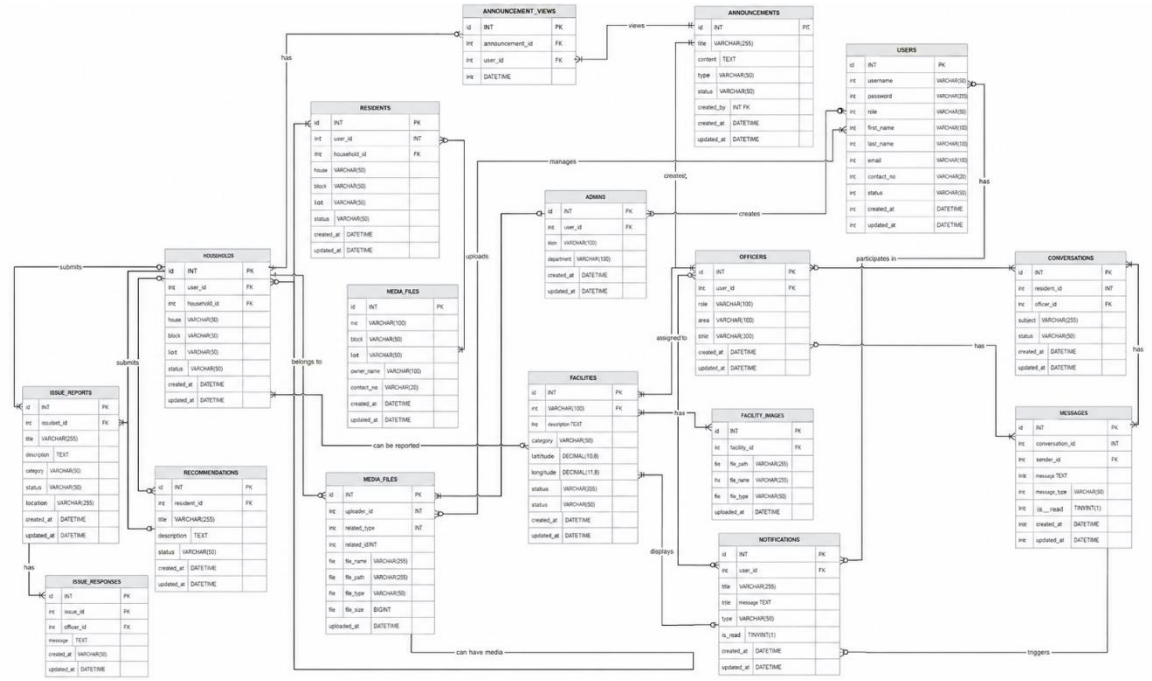
The Entity Relationship Diagram (ERD) for the SubSync system provides a detailed blueprint of the database architecture, illustrating the logical connections and data dependencies between community members and administrative modules. As shown, the

database is centered around three primary user entities: RESIDENT or USER, ADMIN, and OFFICER, each with specific attributes for identification, contact information, and account status. The core of the subdivision's organizational structure is represented by the HOUSEHOLD and HOUSEHOLD_MEMBER tables, which establish a link between residents and their respective living units. To manage community engagement, the ANNOUNCEMENT table stores broadcasted information created by admins or officers, with an ANNOUNCEMENT_VIEW junction table tracking resident interaction to support effective communication across the subdivision.

Operational tasks are further normalized through specialized entities that handle specific community functions. The ISSUE_REPORT table captures resident concerns, including descriptions and relevant location information, while the ISSUE_RESPONSE table links these reports to officers who provide status updates and resolutions. Communication within the platform is facilitated by the CONVERSATION and MESSAGE entities, allowing for direct and organized interactions among users. Furthermore, the FACILITY and MEDIA_FILES tables manage subdivision facilities and uploaded documentation, respectively, while the RECOMMENDATIONS table stores suggestions submitted by residents for community improvement. The NOTIFICATIONS table manages system notifications delivered to users. Together, these entities ensure that subdivision data is systematically stored and easily retrievable to support informed decision-making, effective communication, issue management, and efficient community governance.

Figure 5

Entity Relationship Diagram



Use Case Diagram

A use case diagram is a visual guide that shows how users interact with a system and the actions they can perform. It helps explain the main features and functions of a system in a simple and organized way by identifying the users, also called actors, and the tasks they can perform within the system. This type of diagram is important because it gives developers, researchers, and stakeholders a clear understanding of how the system operates and how each user will interact with it. It also helps ensure that user requirements are considered during the planning and development of the system (GeeksforGeeks, 2025; EDUCBA, 2023).

Figure 5 presents the use case diagram of SubSync: An Automated Web-Based Platform for Enhancing Community Engagement in Residential Subdivisions. The diagram illustrates the interactions among the four main actors of the system: Homeowner, Resident, Officer, and Admin.

The Homeowner can register an account, log in to the system, update profile information, submit complaints or issue reports, view announcements, and receive system notifications. These functions allow homeowners to actively participate in community activities and communicate concerns efficiently.

The Resident can manage household information and household members, view the status of submitted concerns, access announcements, review complaints and reports, receive notifications, and view available reports. These functions help residents stay informed and engaged with subdivision activities.

The Officer is responsible for supporting community operations by creating announcements, managing community policies, assigning administrative responsibilities, generating reports, and receiving notifications. Officers also use the system to coordinate with administrators and residents in addressing community concerns.

The Admin oversees the overall management of the platform by logging in as an administrator, managing users, managing households and residential records, assigning officers, creating announcements, generating reports, and receiving system notifications.

These functions enable the administrator to maintain system operations, monitor community activities, and ensure the efficient administration of the residential subdivision.

Overall, the use case diagram demonstrates how SubSync promotes effective communication, issue management, household administration, and community engagement by providing each user with role-specific functions within a centralized web-based platform.

Figure 6

Use Case Diagram



Data Dictionary

Table 3

Admin

Field Name	Data Type	Length	Description
Admin_ID	int	-	Unique identifier for each admin record.
Name	string	100	Full name of the administrator.
Email	string	150	Email address used for admin login and communication.
Password	string	255	Hashed password used for admin authentication.
Contact_Number	string	20	Contact phone number of the administrator.
Status	string	50	Current account status (e.g., Active, Inactive).
created_at	datetime	-	Date and time when the admin record was created.
updated_at	datetime	-	Date and time of the most recent update to the record.

Stores administrator accounts responsible for managing the barangay system, including user management, financial records, announcements, facility oversight, and overall platform administration.

Table 4

Officer

Field Name	Data Type	Length	Description
Officer_ID	int	-	Unique identifier for each officer record.
Name	string	100	Full name of the barangay officer.
Email	string	150	Email address used for officer login and communication.
Password	string	255	Hashed password used for officer authentication.
Contact_Number	string	20	Contact phone number of the officer.
Status	string	50	Current account status (e.g., Active, Inactive).
Role_Description	string	255	Description of the officer's specific role or designation.

Field Name	Data Type	Length	Description
created_at	datetime	-	Date and time when the officer record was created.
updated_at	datetime	-	Date and time of the most recent update to the record.

Stores barangay officer accounts who are responsible for handling issue responses, posting announcements, managing conversations with residents, and assisting in map management.

Table 5

Resident

Field Name	Data Type	Length	Description
Resident_ID	int	-	Unique identifier for each resident record.
House_ID	int	-	Foreign key referencing the household the resident belongs to.
Name	string	100	Full name of the resident.

Field Name	Data Type	Length	Description
Email	string	150	Email address used for resident login and communication.
Password	string	255	Hashed password used for resident authentication.
Contact_Number	string	20	Contact phone number of the resident.
status	string	50	Current account status (e.g., Active, Inactive).
Current_Balance	decimal	-	Resident's current financial balance within the barangay system.
created_at	datetime	-	Date and time when the resident record was created.
updated_at	datetime	-	Date and time of the most recent update to the record.

Stores resident accounts who interact with the platform to submit issue reports, recommendations, view announcements, monitor financial balances, and communicate with barangay officials.

Table 6

Household

Field Name	Data Type	Length	Description
House_ID	int	-	Unique identifier for each household record.
Block_Lot_Number	string	50	Block and lot number assigned to the property.
statusStatus	string	50	Current occupancy status of the property (e.g., Occupied, Vacant).
Latitude	decimal	-	Geographic latitude coordinate for map geolocation.
Longitude	decimal	-	Geographic longitude coordinate for map geolocation.
created_at	decimal	-	Date and time when the household record was created.

Stores information about each household or property registered within the barangay, including geospatial coordinates used for map display and location tracking.

Table 7

Household_member

Field Name	Data Type	Length	Description
Member_ID	int	-	Unique identifier for each household member record.
House_ID	int	-	Foreign key referencing the household the member belongs to.
Name	string	100	Full name of the household member.
Relationship	string	50	Relationship of the member to the resident (e.g., Spouse, Child, Sibling).
Contact_Number	string	20	Contact phone number of the household member.
created_at	datetime	-	Date and time when the household member record was created.

Stores information about individuals residing within a household who are associated with a registered household but do not have their own system account.

Table 8

Announcement

Field Name	Data Type	Length	Description
Announcement_ID	int	-	Unique identifier for each announcement record.
Admin_ID	int	-	Foreign key referencing the admin who posted the announcement.
Officer_ID	int	-	Foreign key referencing the officer who posted the announcement.
Title	string	200	Short and descriptive title of the announcement.
Content	text	-	Full body content of the announcement.
created_at	datetime	-	Date and time when the announcement was created.
updated_at	datetime	255	Date and time when the announcement was created.

Stores barangay announcements created and posted by administrators or officers, which are made available for residents to view through the platform.

Table 9

Announcement_view

Field Name	Data Type	Length	Description
View_ID	int	-	Unique identifier for each announcement view record.
Resident_ID	int	-	Foreign key referencing the resident who viewed the announcement.
Announcement_ID	int	10,2	Foreign key referencing the announcement that was viewed.
created_at	datetime	255	Date and time when the announcement was viewed by the resident.

Tracks and records which residents have viewed specific announcements, serving as a junction table between the Resident and Announcement entities.

Table 10

Media

Field Name	Data Type	Length	Description
Media_ID	int	-	Unique identifier for each media record.
Uploader_ID	int	-	Identifier of the entity who uploaded the file.
Officer_ID	int	-	Foreign key referencing the officer uploader, if applicable.
Admin_ID	int	-	Foreign key referencing the admin uploader, if applicable.
Resident_ID	int	-	Foreign key referencing the resident uploader, if applicable.
Uploader_Type	string	50	Type of the uploader entity (e.g., Admin, Officer, Resident).
Entity_Type	string	50	Type of entity the media is associated with (e.g., Announcement, Issue).
Entity_ID	int	-	Identifier of the associated entity (polymorphic reference).

Field Name	Data Type	Length	Description
File_Name	string	255	Original file name of the uploaded media.
File_Path	string	500	Storage path or URL where the file is saved.
created_at	datetime		Date and time when the media file was uploaded.

Stores uploaded media files and attachments associated with various entities in the system. This table uses a polymorphic association, allowing files to be linked to announcements, issue reports, residents, officers, or admins.

Table 11

Issue Report

Field Name	Data Type	Length	Description
Issue_ID	int	-	Unique identifier for each issue report record.
Resident_ID	int	-	Foreign key referencing the resident who submitted the report.
Category	string	100	Category of the issue (e.g., Infrastructure,

			Safety, Sanitation).
Title	string	200	Short descriptive title of the issue report.
Description	text	-	Detailed description of the reported issue.
Latitude	decimal	-	Geographic latitude of the location where the issue was reported.
Longitude	decimal		Geographic longitude of the location where the issue was reported.
Status	string	50	Current status of the issue (e.g., Pending, In Progress, Resolved).

Field Name	**Data Type**	**Length**	**Description**
created_at	datetime	-	Date and time when the issue report was submitted.
updated_at	datetime	-	Date and time of the most recent update to the issue report.

Stores barangay issue reports submitted by residents. Each report includes a category, description, and geospatial location data to support map-based tracking and monitoring of community concerns.

Table 12

Issue_response

Field Name	Data Type	Length	Description
Response_ID	int	-	Unique identifier for each issue response record.
Issue_ID	int	-	Foreign key referencing the issue report being responded to.
Officer_ID	int	-	Foreign key referencing the officer who provided the response, if applicable.
Admin_ID	int	-	Foreign key referencing the admin who provided the response, if applicable.
Responder_Type	string	50	Indicates whether the responder is an Officer or an Admin.
Response_Content	datetime	-	Full content of the response or action taken on the issue.
created_at	datetime	-	Date and time when the response was submitted.

Stores official responses to submitted issue reports, provided by barangay officers or administrators. Records the responder type and the content of the action or reply given.

Table 13

Reccomendation

Field Name	Data Type	Length	Description
Recommendation_ID	int	-	Unique identifier for each recommendation record.
Resident_ID	int	-	Foreign key referencing the resident who submitted the recommendationnumber assigned to the payment transaction.
Title	string	200	Short descriptive title of the recommendation.
Description	text	-	Detailed content of the suggestion or recommendation.
Status	string	50	Current status of the recommendation (e.g., Pending, Reviewed, Approved).
created_at	datetime		Date and time when the recommendation

	was submitted.

Stores suggestions and recommendations submitted by residents to the barangay administration. These are reviewed and managed by administrators or officers for potential implementation.

Table 14

Conversation

Field Name	Data Type	Length	Description
Conversation_ID	int	-	Unique identifier for each conversation thread.
Title	string	200	Optional subject or title of the conversation thread.
created_at	datetime	-	Date and time when the conversation thread was created.

Stores conversation threads initiated between residents and barangay officers or admins within the platform's messaging system.

Table 15

Conv_Participant

Field Name	Data Type	Length	Description

Conversation_ID	int	-	Foreign key referencing the associated conversation thread.
Resident_ID	int	-	Foreign key referencing the resident participant, if applicable.
Officer_ID	int	-	Foreign key referencing the officer participant, if applicable.

Field Name	**Data Type**	**Length**	**Description**
Participant_Type	string	50	Foreign key referencing the user who availed the loan.
created_at	datetime	-	Foreign key referencing the type of credit/loan product.

A junction table that defines which residents and officers are participants within a given conversation thread, supporting the messaging system's multi-participant structure.

Table 16

Massage

Field Name	**Data Type**	**Length**	**Description**
Message_ID	int	-	Unique identifier for each message record.
Conversation_ID	int	-	Foreign key

			referencing the conversation the message belongs to.
Resident_ID	int	-	Foreign key referencing the resident sender, if applicable.
Officer_ID	int	-	Foreign key referencing the officer sender, if applicable.
Sender_Type	string	50	Indicates whether the sender is a Resident or an Officer.
Content	text	-	Full text content of the message.
created_at	datetime	-	Date and time when the message was sent.

Stores individual messages sent within a conversation thread by either a resident or an officer, forming the communication history of each conversation.

Table 17

Facility

Field Name	Data Type	Length
Facility_ID	int	-
Admin_ID	int	-

Name	string	150

Field Name	Data Type	Length
Description	string	500
Latitude	decimal	-
Longitude	decimal	-
Status	string	50
created_at	datetime	-

Stores information about barangay facilities managed by administrators, including geospatial coordinates for display on the community map and operational status tracking.

Recommended Hardware Requirements

To ensure the smooth operation, scalability, and optimal performance of the SubSync system especially considering its media handling and geospatial mapping features the following upgraded hardware specifications are recommended.

Web Server

Component	Specification
Processor (CPU)	Multi-core processor (4 to 8 cores), 2.5 GHz or higher
Memory (RAM)	16 GB – 32 GB (to efficiently handle concurrent database queries and mapping data)
Storage	16 GB – 32 GB (to efficiently handle concurrent database queries and mapping Intel Core i3 / AMD Ryzen 3 equivalent or higherdata)

Admin/Officer Workstation (Client PC)

Component	Specification
Processor (CPU)	Intel Core i3 / AMD Ryzen 3 equivalent or higher
Memory (RAM)	8 GB
Display	1920 × 1080 resolution recommended (for clear visibility of the facility map and data tables)

Mobile Application (Resident Device)

Component	Specification
Memory (RAM)	3 GB or higher
Storage	100 MB free space for app installation and cache
Processor	64-bit ARM-based processor

Sensors & Hardware	GPS (for location tracking on issue reports), Camera (for media uploads), and active Internet connection

Software Requirements

To ensure secure and efficient access to the SubSync platform, the following software specifications are recommended..

Web Server & Database Environment

Component	Specification
Operating System	Ubuntu 22.04 LTS, Debian 11, or Windows Server 2022
Database Engine	MySQL 8.0, PostgreSQL 14, or higher

Web Admin Dashboard (Client-side)

Component	Specification
Operating System	Windows 10 / Windows 11 / macOS 12+ / Modern Linux distributions
Web Browser	Latest stable release of Google Chrome, Mozilla Firefox, Apple Safari, or Microsoft Edge

Mobile Application

Component	Specification
Android OS	Android 8.0 (Oreo / API 26) or newer
iOS	iOS 13.0 or newer

GANTT CHART

The Gantt Chart is used by the proponents of the study as a tool to present the sequence of activities, schedule, and specific tasks necessary for the development of SubSync: An Automated Web-Based Platform for Enhancing Community Engagement in Residential Subdivisions. It outlines the major phases of the study, including planning, requirements analysis, system design, development, testing, evaluation, and documentation, together with the estimated duration and target dates for each activity.

The Gantt chart serves as a guide for monitoring the progress of the project and ensuring that each task is completed within the proposed timeframe. It also helps the researchers organize the iterative development process based on the Agile methodology applied in the study. Furthermore, the chart includes allowances for revisions, debugging, consultations, and unforeseen delays that may occur during the implementation and evaluation of the system. Through this timeline, the proponents can effectively manage project activities and maintain the systematic development of the SubSync platform.

SubSync: An Automated Web-Based Platform for Enhancing Community Engagement in Residential Subdivisions																
Activity			February		March				April				May		
			3rd Week	4th Week	1st Week	2nd Week	3rd Week	4th Week	1st Week	2nd Week	3rd Week	4th Week	1st Week	2nd Week	3rd week
Iteration 1		**Assigned Member:**													
1.1 Initiation	Planning	ALL													
1.2 Initiation	Finalizing Capstone Title	ALL													
1.3 Planning	Title Hearing	ALL													
1.4 Planning	Making and Signing of Letters	ALL													
1.5 Analysis	Site Visit and Beneficiary Approval	TIBO-OC													
1.6 Analysis	Submission of Letters	ALL													
1.7 Design	System Branding Design and UI Design	TIBO-OC													
1.8 Documentation	Finalizing Chapter 1 and 2	SALAZAR & GUANZON													
1.9 Documentation	Chapter 1 Submission	SALAZAR													
2.1 Documentation	Chapter 2 Submission	SALAZAR													
Iteration 2															
2.2 Development	System Development	ALL													
2.3 Testing	Progress Update Submission	SALAZAR													
2.4 Evaluation	Site Validation	ALL													
2.5 Development	Authentication & User Management Module	BARBAS													
2.6 Development	Payment Tracking and Communication Module	BARBAS													
2.7 Development	Issue Reporting and Announcement Module	BARBAS & TIBO-OC													
2.8 Development	Dynamic Mapping / GIS Integration	BARBAS													
2.9 Development	Administrative Dashboard & Analytics	TIBO-OC & GUANZON													
3.0 Testing	Unit / Functional Testing	ALL													
3.1 Evaluation	Final Revisions for Chapter 3	ALL													

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APPENDICES

APPENDIX A

Carlos Hilado Memorial State University
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College of Computer Studies

IT CAPSTONEPROJECT ADVISER ACCEPTANCE FORM

Date: March 12, 2026

This is to attest that I, **Dexter P. Hormigoso**, have agreed and accepted to be the **Adviser** of the following students with the Systems Project entitled ***"SubSync: An Automated Web-Based Platform for Enhancing Community Engagement in Residential Subdivisions"***.

Group Name: **CalmB4DStorm**
Course/Yr./Sec: **BSIT 3B**

Members:

1. **Christian J. Barbas Jr.**
2. **Jurriel A. Guanzon**
3. **Shirley Ann M. Salazar**
4. **Paul Felippe G. Tibo-oc**

Furthermore, I agree to set the schedule for advising and consultation to help the students and ensure the success of the project.

Conformed:

DEXTER P. HORMIGOSO, LPT, MIT
Adviser's Signature over Printed Name

Noted by:

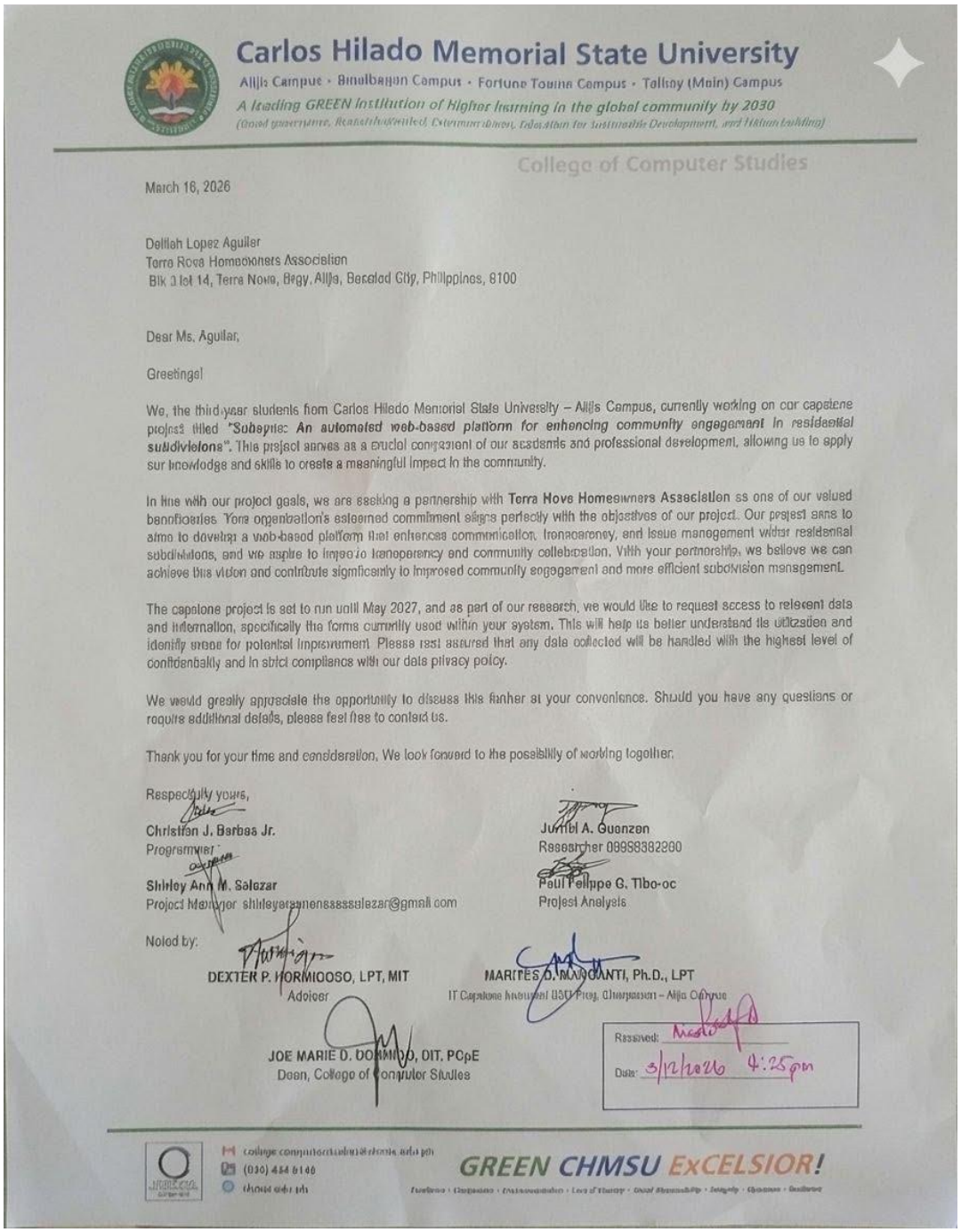
MARITES D. MANGANTI, LPT, Ph.D.
IT Capstone Project Professor

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APPENDIX B

Letter for Beneficiary



Carlos Hilado Memorial State University
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March 16, 2026

Delilah Lopez Aguilar
Terra Rosa Homeowners Association
Blk 3 lot 14, Terra Nova, Brgy. Alijis, Bacolod City, Philippines, 6100

Dear Ms. Aguilar,

Greetings!

We, the third-year students from Carlos Hilado Memorial State University – Alijis Campus, currently working on our capstone project titled "**Subaybay: An automated web-based platform for enhancing community engagement in residential subdivisions**". This project serves as a crucial component of our academic and professional development, allowing us to apply our knowledge and skills to create a meaningful impact in the community.

In line with our project goals, we are seeking a partnership with **Terra Nova Homeowners Association** as one of our valued beneficiaries. Your organization's esteemed commitment aligns perfectly with the objectives of our project. Our project aims to aims to develop a web-based platform that enhances communication, transparency, and issue management within residential subdivisions, and we aspire to implement transparency and community collaboration. With your partnership, we believe we can achieve this vision and contribute significantly to improved community engagement and more efficient subdivision management.

The capstone project is set to run until May 2027, and as part of our research, we would like to request access to relevant data and information, specifically the forms currently used within your system. This will help us better understand its utilization and identify areas for potential improvement. Please rest assured that any data collected will be handled with the highest level of confidentiality and in strict compliance with our data privacy policy.

We would greatly appreciate the opportunity to discuss this further at your convenience. Should you have any questions or require additional details, please feel free to contact us.

Thank you for your time and consideration. We look forward to the possibility of working together.

Respectfully yours,

Christian J. Barbas Jr.
Programmer

Shirley Ann M. Salazar
Project Manager shirleyannmsalazar@gmail.com

Junibi A. Guanzon
Researcher 09958382280

Paul Philippe G. Tibo-oc
Project Analysis

Noted by:

DEXTER P. HORMIGOSO, LPT, MIT
Adviser

MARITES D. MANGANTI, Ph.D., LPT
IT Capstone Research Unit Proj. Chairperson – Alijis Campus

JOE MARIE D. DOMINGO, DIT, PCpE
Dean, College of Computer Studies

Received: Mario
Date: 3/17/2026 4:25pm

college.computerstudies@chmsu.edu.ph
(034) 434 8146
chmsu.edu.ph

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Beneficiary Acknowledgement



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BENEFICIARY ACKNOWLEDGMENT AND ACCEPTANCE CERTIFICATE

This is to certify that **Delilah Lopez Aguilar**, representing Terra Nova Homeowners Association, acknowledges and accepts being the official beneficiary of the capstone project titled **"Subsync: An automated web-based platform for enhancing community engagement in residential subdivisions ".**

The capstone project, conducted by students from **Carlos Hilado Memorial State University – Alijis Campus** aims to provide innovative solutions aligned with the needs and objectives of the beneficiary. As part of this collaboration, the beneficiary agrees to provide necessary support, feedback, and relevant information to facilitate the successful completion of the project.

By signing this certificate, the beneficiary expresses their consent and commitment to participate in the study and acknowledges the importance of mutual cooperation for the benefit of both parties.

Issued this 12th day of March, 2026 at Carlos Hilado Memorial State University – Alijis Campus, Bacolod City, Negros Occidental.

Beneficiary Details:

Signature Over Printed Name/ Date

Designation: President

Organization: Terra Nova Homeowners Association

Students Details:

christian J. Barbas Jr. | JURRAEL A. GUANZON | SALAZAR, SHIRLEY ANN M.

Signature Over Printed Name/ Date | Signature Over Printed Name/ Date | Signature Over Printed Name/ Date

PAUL FELIPPE G. TIBO-OC

Signature Over Printed Name/ Date

Attested by:

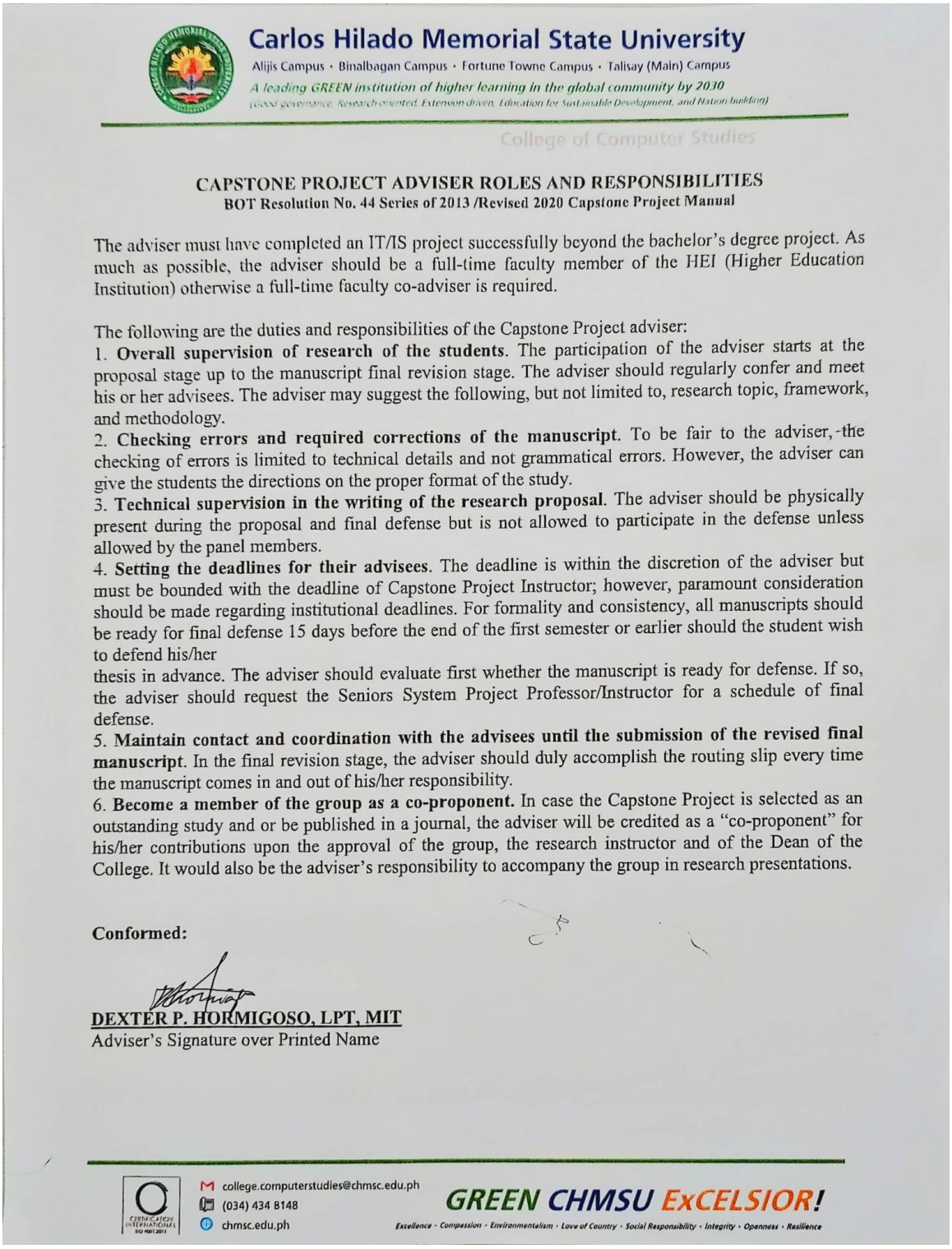
DEXTER P. HORMIGOSO, LPT, MIT
Adviser

MARITES D. MANGANTI, LPT, Ph.D.
Capstone Project Instructor

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CAPSTONE PROJECT ADVISER ROLES AND RESPONSIBILITIES
BOT Resolution No. 44 Series of 2013 /Revised 2020 Capstone Project Manual

The adviser must have completed an IT/IS project successfully beyond the bachelor's degree project. As much as possible, the adviser should be a full-time faculty member of the HEI (Higher Education Institution) otherwise a full-time faculty co-adviser is required.

The following are the duties and responsibilities of the Capstone Project adviser:

1. **Overall supervision of research of the students**. The participation of the adviser starts at the proposal stage up to the manuscript final revision stage. The adviser should regularly confer and meet his or her advisees. The adviser may suggest the following, but not limited to, research topic, framework, and methodology.
2. **Checking errors and required corrections of the manuscript**. To be fair to the adviser, the checking of errors is limited to technical details and not grammatical errors. However, the adviser can give the students the directions on the proper format of the study.
3. **Technical supervision in the writing of the research proposal**. The adviser should be physically present during the proposal and final defense but is not allowed to participate in the defense unless allowed by the panel members.
4. **Setting the deadlines for their advisees**. The deadline is within the discretion of the adviser but must be bounded with the deadline of Capstone Project Instructor; however, paramount consideration should be made regarding institutional deadlines. For formality and consistency, all manuscripts should be ready for final defense 15 days before the end of the first semester or earlier should the student wish to defend his/her
thesis in advance. The adviser should evaluate first whether the manuscript is ready for defense. If so, the adviser should request the Seniors System Project Professor/Instructor for a schedule of final defense.
5. **Maintain contact and coordination with the advisees until the submission of the revised final manuscript**. In the final revision stage, the adviser should duly accomplish the routing slip every time the manuscript comes in and out of his/her responsibility.
6. **Become a member of the group as a co-proponent.** In case the Capstone Project is selected as an outstanding study and or be published in a journal, the adviser will be credited as a "co-proponent" for his/her contributions upon the approval of the group, the research instructor and of the Dean of the College. It would also be the adviser's responsibility to accompany the group in research presentations.

Conformed:

DEXTER P. HORMIGOSO, LPT, MIT
Adviser's Signature over Printed Name

college.computerstudies@chmsc.edu.ph
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Evaluation Tools

RESEARCH INSTRUMENT

SubSync: An Automated Web-Based Platform for Enhancing Community Engagement in Residential Subdivisions

Type of Respondent :

☐ **Subdivision Administrator** ☐ **HOA Officer** ☐ **Resident**

☐ **IT Expert / IT Professional**

Direction: The following statements are designed to assess the software product quality of the SubSync System based on the ISO/IEC 25010:2011 Software Quality Standard. Kindly review each statement carefully and place a **check (✓)** in the box that corresponds to your answer using the rating scale below.

Rating scale:

Code	Interpretation	Description
5	Excellent	4.51 – 5.00. The system is complete and fully functional.
4	Very Good	3.51 – 4.50. The system is complete and functional.
3	Good	2.51 – 3.50 The system is somewhat complete and functional.
2	Poor	1.51 – 2.50. The system is incomplete and somewhat functional.
1	Very Poor	1.50 - below. The system is incomplete and non-functional.

A. Functional Suitability	5	4	3	2	1

4. **Functional completeness.** Degree to which the set of functions covers all the specified tasks and user objectives.					
5. **Functional correctness.** Degree to which a product or system provides the correct results with the needed degree of precision.					
6. **Functional appropriateness.** Degree to which the functions facilitate the accomplishment of specified tasks and objectives.					
Total Score / Mean Score					

B. Performance Efficiency	**5**	**4**	**3**	**2**	**1**
a. **Time Behavior.** The response and processing times and throughput rates of a product or system, when performing its functions meet requirements.					
b. **Resource Utilization.** The amount and types of resources used by the system, when performing its functions, meet requirements.					
c. **Capacity.** The maximum limits of a product or system parameter meet requirements.					
Total Score / Mean Score					

C. **Usability**	**5**	**4**	**3**	**2**	**1**
1. **Appropriateness recognizability.** Degree to which users can recognize whether a product or system is appropriate for their needs.					
2. **Learnability.** degree to which a product or system can be used by specified users to achieve specified goals of learning to use the product or system with					

effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use.					
3. **Operability.** Degree to which a product or system has attributes that make it easy to operate and control.					
4. **User error protection.** Degree to which a system protects users against making errors.					
5. **User interface aesthetics.** Degree to which a user interface enables pleasing and satisfying interaction for the user.					
6. **Accessibility.** Degree to which a product or system can be used by people with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use.					
Total Score / Mean Score					

D. Compatibility	**5**	**4**	**3**	**2**	**1**
1. **Co-existence.** Degree to which a product can perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product.					
2. **Interoperability**. Degree to which two or more systems, products or components can exchange information and use the information that has been exchanged.					
Total Score / Mean Score					

E. Reliability	**5**	**4**	**3**	**2**	**1**

1. **Maturity.** Degree to which a system, product or component meets needs for reliability under normal operation.					
2. **Availability.** Degree to which a system, product or component is operational and accessible when required for use.					
3. **Fault tolerance.** Degree to which a system, product or component operates as intended despite the presence of hardware or software fault					
4. **Recoverability.** Degree to which, in the event of an interruption or a failure, a product or system can recover the data directly affected and re-establish the desired state of the system.					
Total Score / Mean Score					

F. Security	**5**	**4**	**3**	**2**	**1**
1. **Confidentiality.** Degree to which a product or system ensures that data are accessible only to those authorized to have access.					
2. **Integrity.** Degree to which a system, product or component prevents unauthorized access to, or modification of, computer programs or data.					
3. **Non-repudiation.** degree to which actions or events can be proven to have taken place, so that the events or actions cannot be repudiated later.					
4. **Accountability.** Degree to which the actions of an entity can be traced uniquely to the entity.					
5. **Authenticity.** Degree to which the identity of a subject or					

resource can be proved to be the one claimed.					
Total Score / Mean Score					

G. Maintainability	**5**	**4**	**3**	**2**	**1**
1. **Modularity.** Degree to which a system or computer program is composed of discrete components such that a change to one component has minimal impact on other components.					
2. **Reusability.** Degree to which an asset can be used in more than one system, or in building other assets.					
3. **Analysability.** Degree of effectiveness and efficiency with which it is possible to assess the impact on a product or system of an intended change to one or more of its parts, or to diagnose a product for deficiencies or causes of failures, or to identify parts to be modified.					
4. **Modifiability.** Degree to which a product or system can be effectively and efficiently modified without introducing defects or degrading existing product quality.					
5. **Testability.** Degree of effectiveness and efficiency with which test criteria can be established for a system, product or component and tests can be performed to determine whether those criteria have been met.					
Total Score / Mean Score					

H. Portability	5	4	3	2	1
1. **Adaptability.** Degree to which a product or system can effectively and efficiently be adapted for different or evolving hardware, software or other operational or usage environments.					
2. **Installability.** Degree of effectiveness and efficiency with which a product or system can be successfully installed and/or uninstalled in a specified environment.					
3. **Replaceability.** Degree to which a product can replace another specified software product for the same purpose in the same environment.					
Total Score / Mean Score					
Overall Score and Mean Score					

_____

Signature

_____

Date

TEST CASE EVALUATION INSTRUMENT FOR IT PROFESSIONALS

Evaluator Name (Optional): _____
Age: _____ **Sex:** _____
Occupation: _____
Device Used: _____
Experience Level: _____

GENERAL INSTRUCTIONS: This test case evaluation checklist is designed to verify the functional correctness of the **SubSync Platform**. As an IT Professional, you are requested to execute the predefined test steps and verify if the system's actual behavior matches the expected result. Please mark the status as **"Pass"** if the system functions correctly, or **"Fail"** if an error or unexpected behavior occurs. You may use the remarks column to note technical observations, bugs, or mapping/algorithmic inaccuracies.

I. Test Case Rating Scale and Interpretation

Percentage of Passed Test Case	Descriptive Rating	Interpretation
96% – 100%	Excellent	The system passed almost all test cases and functions correctly with minimal or no errors
90% – 95%	Very Good	The system passed most test cases with only minor issues detected.
85% – 89%	Good	The system functions properly but requires some improvements in certain features.
80% – 84%	Fair	The system meets basic functional requirements but several issues need correction.
Below 80%	Poor	The system failed significant test cases and requires major revisions and debugging.

II. Test Case Execution Checklist

Test Case ID	Test Scenario / Objective	Test Description	Preconditions	Test Steps	Expected Result	Status (Pass/Fail)	Remarks
TC-01	User Authentication	Verify login using valid credentials	User account is registered	1. Enter valid username and password 2. Click Login	User is successfully authenticated and redirected to dashboard		
TC-02	Announcement Module	Verify admin can post announcements	Admin account is logged in	1. Navigate to Announcement Module 2. Create an announcement 3. Click Post	Announcement is successfully published and visible to residents		
TC-03	Issue Reporting	Verify residents can submit issue reports	Resident account is logged in	1. Navigate to Issue Reporting 2. Enter issue details 3. Submit report. Open payment history without records	Issue report is successfully saved and visible to authorized administrators/officers		
TC-04	Issue Monitoring	Verify officers/admins can respond to issue reports	Officer/Admin is logged in and an issue report exists	1. Open submitted issue 2. Add response or update 3. Save response	Response/update is successfully saved and the issue status is updated		

TC-05	Recommendation Module	Verify residents can submit recommendations	Resident account is logged in	1. Open Recommendation Page 2. Enter suggestion 3. Submit	Recommendation is successfully stored with the appropriate status		
TC-06	Mapping Feature	Verify the community map displays facilities and issue markers	User is logged in with internet access	1. Open Community Map 2. View facility and issue markers	Map loads correctly and displays available facilities and reported issues		
TC-07	Mapping Feature	Verify invalid location data is handled correctly	User is logged in	1. Submit or select invalid map coordinates/location data	System rejects the invalid location and displays an appropriate error message		
TC-08	Messaging Module	Verify direct messaging between resident and officer/admin	Both accounts are active	1. Open Messaging 2. Select recipient 3. Send a message	Message is successfully sent and received by the intended recipient		
TC-09	Media Upload	Verify image/file upload functionality	User account is logged in	1. Open a supported module 2. Upload an image/file 3. Submit	Media file uploads successfully and is stored/displayed properly		

TC-10	Compatibility and Performance	Verify system accessibility and responsiveness across devices and browsers	Supported devices/browsers are available	1. Open the system on desktop and mobile browsers 2. Perform common operations	1. Open the system on desktop and mobile browsers 2. Perform common operations		

Signature over Printed Name of IT Professional Evaluator: _____

Date: _____

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Data Privacy Consent Form

We are committed to protecting your personal data and ensuring that it is processed in compliance with relevant data protection regulations. All information collected will be used solely for the purposes of this research project and will not be shared with any third parties without your explicit consent. Your participation is entirely voluntary, and you have the right to withdraw from the study at any time.

If you are willing to participate, please indicate your consent by signing and returning the attached consent form to us by ***March 12, 2026***. If you have any questions or require further information, please do not hesitate to contact me at ***09489868980*** or via email at shirleyannmenesessalazar@gmail.com

.

Thank you for considering our request. Your contribution will be instrumental in enhancing the effectiveness of the **" SubSync: An Automated Web-Based Platform for Enhancing Community Engagement in Residential Subdivisions."**

Sincerely,

Shirley Ann M. Salazar
Project Manager
Carlos Hilado Memorial State University
09489868980 shirleyannmenesessalazar@gmail.com

Consent Form
I, **Delilah Lopez Aguilar** , hereby grant permission for **Shirley Ann M. Salazar** and **Carlos Hilado Memorial State University** to conduct an interview with me and access relevant data and information related to my use of the **"Subsnyc"** for the purposes of the research project.

I understand that my participation is voluntary and that I can withdraw from the study at any time. I also acknowledge that all data collected will be treated confidentially and in accordance with the data privacy statement provided.

Signature: DELILAH L. AGUILAR
Date: MAY 14, 2026